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SIR Spread Simulation Across Network Topologies

– Complex Networks –

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Abstract

The adoption and decline of mobile applications can be modelled as a spreading process on a social network. In this paper we adapt the classical Susceptible-Infected-Recovered (SIR) epidemic model to describe three user states: susceptible (S, could install the app), infected (I, currently using and spreading it), and recovered (R, deleted it and will not reinstall). We implement and simulate this model computationally across four canonical network topologies: Erdős–Rényi random graphs, Barabási–Albert scale-free networks, Watts–Strogatz small-world networks, regular lattices, and analyse how structural properties such as degree distribution, clustering coefficient, and average path length determine the speed and reach of app adoption. Our experiments ($N = 200$ nodes, 30 independent runs per topology) show that Erdős–Rényi graphs generate the highest peak prevalence (79.8%) and the fastest epidemic (peak at step 9.6), and that lattices substantially impede spreading (peak 32.6%, reached at step 43.6).

Contents

Chapter 1

Introduction

The rapid rise and fall of mobile applications is one of the most visible examples of social contagion in everyday life. An app can go from zero to millions of installs within days, driven entirely by word-of-mouth recommendations between connected users. This behaviour is structurally analogous to the spread of an infectious disease through a population, suggesting that the rich theoretical framework of epidemic modelling can be applied directly to understand and predict app adoption.

A social network can be represented as a graph in which nodes are users and edges represent social ties through which recommendations flow. The key insight, established by Pastor-Satorras and Vespignani (2001) in the context of disease spreading, is that the topology of this graph profoundly influences the dynamics of any spreading process running on it. A hub-dominated scale-free network behaves very differently from a locally connected lattice, even when the per-edge transmission probability is identical.

This paper has three goals. First, we provide a rigorous theoretical introduction to the SIR compartmental model and the four network topologies. Second, we implement a computational simulation of app-spread SIR dynamics. Third, we present and interpret experimental results comparing the four topologies across metrics including peak active-user prevalence, time to peak and total adoption.

The remainder of the paper is structured as follows. Section 2 introduces the theoretical background. Section 3 describes the model and simulation methodology. Section 4 presents results. Section 5 discusses implications, limitations and concludes.

Chapter 2

Theoretical Background

2.1 The SIR Epidemic Model

The SIR model, introduced by Kermack and McKendrick [3], partitions a fixed population of N individuals into three mutually exclusive compartments at every discrete time step t:

- $S(t)$: Susceptible individuals who have not yet adopted the app but could do so.
- $I(t)$: Infected (active) individuals currently using the app and capable of recommending it to neighbours.
- $R(t)$: Recovered individuals who got bored with the app, deleted it, and won't be reinstalling it.

At each time step, every active node independently attempts two events:

- **Spreading:** for each susceptible neighbour j, node i converts j to I with probability β.
- **Recovery:** node i transitions to R with probability γ.

The conservation law $S(t) + I(t) + R(t) = N$ holds for all t. The system reaches a fixed point when $I(t) = 0$; at that point $S(\infty)$ and $R(\infty) = N - S(\infty)$ are determined by the dynamics.

2.2 The Basic Reproduction Number R_0

The central quantity governing whether a spreading process becomes epidemic is the basic reproduction number [5]:

$$R_0 = \frac{\beta\langle k\rangle}{\gamma} \tag{2.1}$$

where $\langle k \rangle$ is the mean degree of the network. When $R_0 > 1$, each active node infects on average more than one susceptible neighbour before recovering, and a large-scale epidemic is possible. When $R_0 < 1$, the infection dies out exponentially. The critical threshold $R_0 = 1$ separates these regimes.

Crucially, Pastor-Satorras and Vespignani [6] showed that on scale-free networks the effective epidemic threshold vanishes in the thermodynamic limit ($N \to \infty$), because hubs with very high degree $k \gg \langle k \rangle$ can sustain spreading even when R_0 computed from the mean degree would predict extinction. This is one of the landmark results in network epidemiology and has direct implications for app virality: a single influential hub user can ignite a global adoption cascade regardless of how slowly the average user spreads the app.

It is useful to distinguish two related quantities. The ratio β/γ measures the expected number of transmission attempts per edge before a node recovers, and is a property of the disease (or app) alone. The reproduction number $R_0 = \beta\langle k \rangle/\gamma$ additionally incorporates the mean number of contacts a node has, and is therefore a joint property of the disease and the network. In this study we fix $\beta/\gamma = 2.40$ across all simulations so that any differences in outcomes are attributable purely to network structure; the resulting R_0 values differ by topology and are reported alongside the simulation results.

2.3 Network Topologies

We study four topologies that together span the space of real-world social network properties.

Erdős–Rényi (ER) random graph. [2] In $G(N, p)$, each of the $\binom{N}{2}$ possible edges is included independently with probability p. The resulting degree distribution is binomial, well approximated by a Poisson distribution with mean $\langle k \rangle = p(N - 1)$ for large N. The graph is homogeneous: most nodes have degree close to $\langle k \rangle$, and there are no hubs. ER graphs exhibit short average path lengths ($\propto \ln N/ \ln\langle k \rangle$) but low clustering.

Barabási–Albert (BA) scale-free network. [1] Constructed by preferential attachment: starting from a small seed graph, each new node added to the network connects to m existing nodes with probability proportional to their current degree. This ‘rich-get-richer’ mechanism produces a power-law degree distribution $P(k) \propto k^{-3}$, meaning a small number of hubs accumulate very high degree. Scale-free networks are robust to random failure but highly vulnerable to targeted removal of hubs. Average path lengths are ultra-small ($\propto \ln \ln N$).

Watts–Strogatz (WS) small-world network. [7] Constructed by starting from a k-regular ring lattice and rewiring each edge with probability p to a uniformly random node. For intermediate p, the resulting graph retains high local clustering (from the regular structure) while acquiring dramatically shorter path lengths (from the long-range shortcuts). This combination—high C, small L—is the defining feature of small-world networks and is observed in many real social networks [4].

Square Regular lattice (Grid graph). A 2D grid in which each interior node has exactly four neighbours (up, down, left, right). The lattice is the most locally connected structure: clustering is zero, average path length scales as $\sqrt{N}$, and there are no long-range connections. Spreading on a lattice proceeds as a spatial wave emanating from the seed and is the slowest of the four.

Chapter 3

Methodology

3.1 Graph Construction

The parameters for building the graphs were chosen to produce comparable graph sizes ($N \approx 200$) while reflecting the defining structural properties of each topology:

- Erdős–Rényi: $N = 200$, $p = 0.05$, yielding $\langle k \rangle \approx 9.4$.
- Barabási–Albert: $N = 200$, $m = 3$ (edges per new node), yielding $\langle k \rangle \approx 5.9$.
- Watts–Strogatz: $N = 200$, $k = 6$ (initial ring neighbours), rewiring probability $p = 0.1$, yielding $\langle k \rangle = 6.0$.
- Lattice: 14×14 grid, $N = 196$, $\langle k \rangle \approx 3.7$ (boundary nodes have degree 2 or 3).

3.2 Simulation Protocol

The procedure for a single run is as follows:

- **Initialisation:** a random fraction $\iota_0 = 2\%$ of nodes are seeded as I; all others start as S.
- **State update:** at each step, all currently I nodes are evaluated. With probability γ a node transitions to R; at the same time, for each susceptible neighbour, the node converts that neighbour to I with probability β. Updates within a step are applied synchronously.
- **Termination:** the simulation runs until $I(t) = 0$ or $t = T_{\max} = 150$.

To account for stochastic variability, each topology was simulated for $n = 30$ independent runs using different random seeds. Results are reported as means with standard deviations across runs.

The fixed parameters throughout are $\beta = 0.12$ and $\gamma = 0.05$. Because $R_0 = \beta\langle k\rangle/\gamma$ depends on network topology through $\langle k\rangle$, it differs across the four graphs: $R_0 \approx 22.6$ (ER), 14.2 (BA), 14.4 (WS), and 8.9 (Lattice). All four values are well above the epidemic threshold $R_0 = 1$, so a large outbreak is theoretically expected in every topology.

3.3 Metrics

We report the following metrics for each topology:

- **Peak active prevalence** I^*/N (%): maximum fraction of simultaneously active users across the run.
- **Time to peak** t^*: the time step at which $I(t)$ is maximised.
- **Final adoption** $R(\infty)/N$ (%): fraction of nodes that ever adopted the app (equivalent to $1 - S(\infty)/N$).
- **Network-science metrics:** average degree $\langle k\rangle$, maximum degree $k_{\max}$, clustering coefficient C, average path length L, and graph diameter d.

Chapter 4

Results

4.1 Network Structure

Table 4.1 reports the empirical network metrics computed from the generated graphs. The results confirm the theoretical expectations. The ER graph has the highest average degree ($\langle k \rangle = 9.42$) because its edge probability $p = 0.05$ was chosen to ensure connectivity, resulting in a denser graph. The BA network has the highest maximum degree ($k_{\max} = 41$, compared to 17 for ER and 8 for WS), reflecting the hub structure produced by preferential attachment. The WS network achieves by far the highest clustering coefficient ($C = 0.438$), while the lattice has the longest average path length ($L = 9.33$) and diameter ($d = 26$), consistent with its $\sqrt{N}$ scaling.

Table 4.1: Empirical network metrics ($N \approx 200$).

Topology	N	**Edges**	$\langle k \rangle$	$k_{\max}$	C	L	d
Erdős–Rényi	200	942	9.42	17	0.046	2.59	4
Barabási–Albert	200	591	5.91	41	0.102	2.83	5
Watts–Strogatz	200	600	6.00	8	0.438	4.42	8
Lattice	196	364	3.71	4	0.000	9.33	26

4.2 SIR Epidemic Dynamics

Figure 4.3 shows the averaged SIR curves (mean ± 1 standard deviation across 30 runs) for each topology, and Figure 4.4 overlays the active-user curves $I(t)/N$ for all four topologies on a single axis for direct comparison.

The results reveal a clear ordering in spreading speed and intensity.

Erdős–Rényi produced the highest and earliest peak: $I^*/N = 79.8\%$ ($\pm 2.8\%$) reached at $t^* = 9.6$

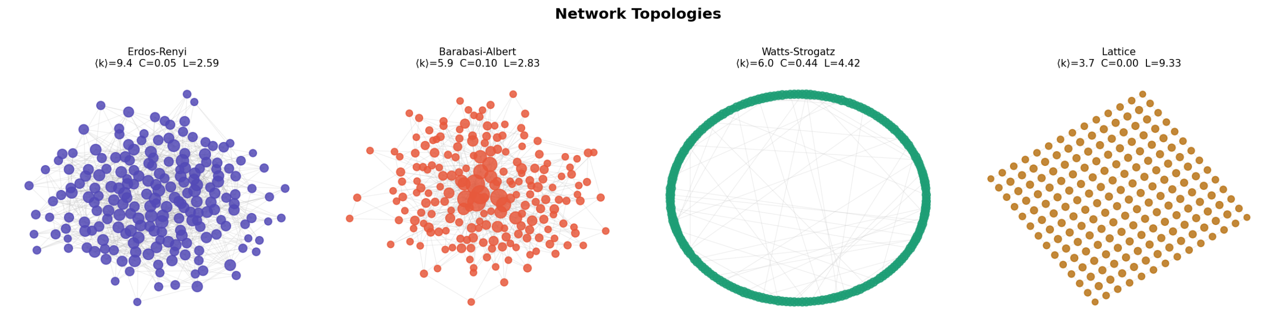

Figure 4.1: Visual representation of the four network topologies ($N = 200$). Node size is proportional to degree. Hubs are clearly visible in the Barabási–Albert graph.

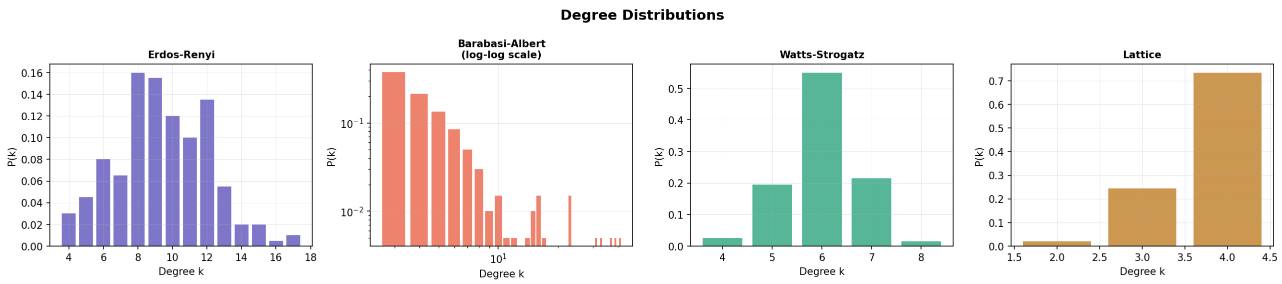

Figure 4.2: Degree distributions for each topology. The Barabási–Albert plot is shown on a log-log scale to reveal the power-law tail $P(k) \propto k^{-3}$.

(± 1.3) steps. The homogeneous degree distribution, combined with the high mean degree ($\langle k \rangle = 9.4$), gives every node a large neighbourhood to infect simultaneously. The epidemic burns through nearly the entire network (final $R(\infty)/N = 100.0\%$) in a narrow time window.

Barabási–Albert achieved a peak of $I^*/N = 69.0\%$ ($\pm 3.7\%$) at $t^* = 11.7$ (± 1.6). Despite having a lower mean degree than ER, the hub nodes ($k_{\max} = 41$) create super-spreading events in the early phase, rapidly seeding the epidemic across the network. Final adoption was 98.5%, very close to saturation.

Watts–Strogatz showed a slower but still large epidemic: $I^*/N = 59.0\%$ ($\pm 4.8\%$) at $t^* = 21.1$ (± 3.0). The high clustering coefficient ($C = 0.438$) means that a node's neighbours tend to already know each other, so once a local cluster is infected the spreading must rely on the sparse long-range shortcuts to reach new territory. This slows the epidemic but does not prevent it—final adoption reached 99.7%.

Lattice produced the slowest and least intense epidemic: $I^*/N = 32.6\%$ ($\pm 6.8\%$) at $t^* = 43.6$ (± 18.6). The absence of long-range connections forces the epidemic to propagate as a spatial wave expanding from the seed, constrained to a maximum speed of one lattice cell per step. The large variability ($\sigma = 6.8\%$ for peak, 14.6% for final adoption) reflects the sensitivity of the wave to the

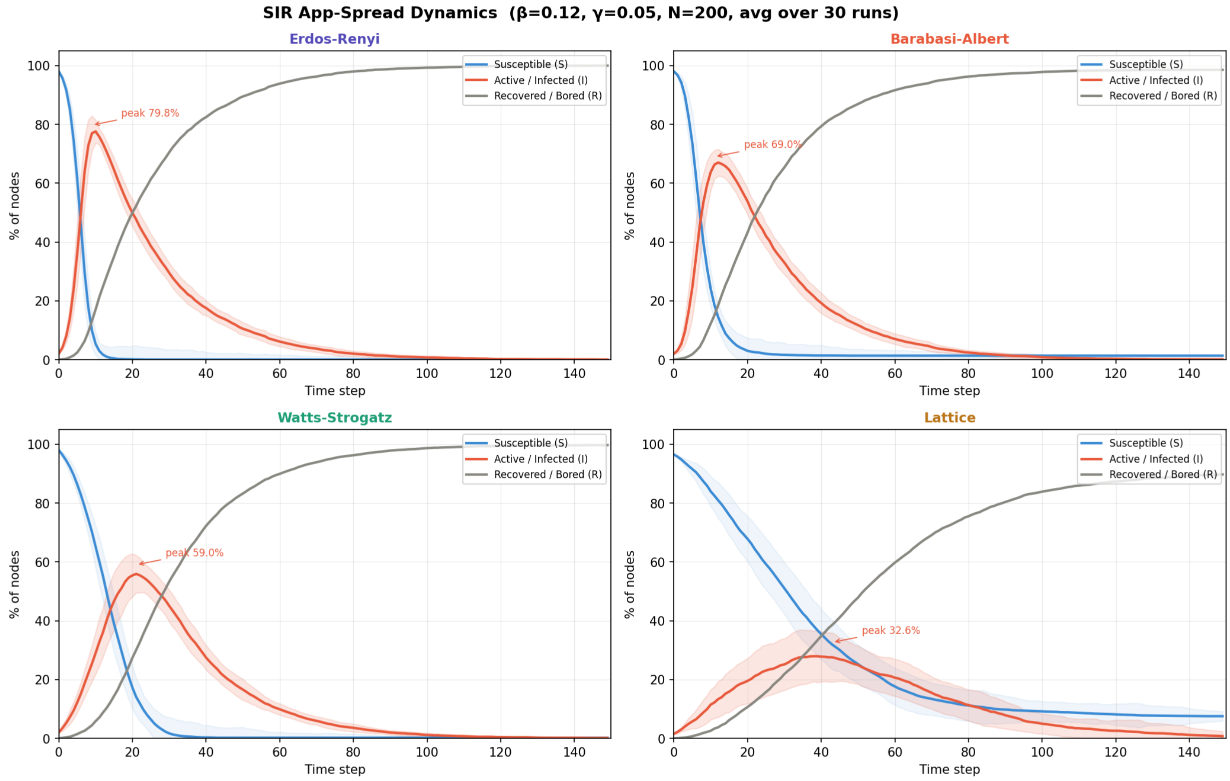

Figure 4.3: Mean SIR curves ($\pm 1\sigma$ shading) for each topology. Parameters: $\beta = 0.12$, $\gamma = 0.05$ (R_0 varies by topology), $N \approx 200$, 30 independent runs.

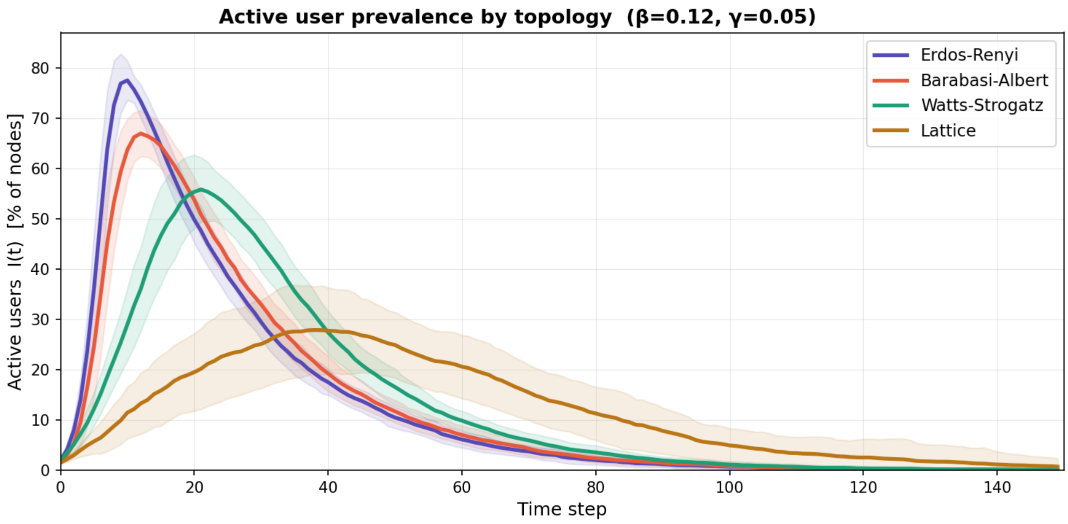

Figure 4.4: Overlay of active-user prevalence $I(t)/N$ across all four topologies. $\beta = 0.12$, $\gamma = 0.05$ $\pm 1\sigma$ shading included. The ordering from fastest to slowest is: Erdős–Rényi > Barabási–Albert > Watts–Strogatz > Lattice.

random initial seed position. In some runs the epidemic failed to reach portions of the grid before dying out, resulting in final adoption as low as 70%.

Table 4.2: Summary of SIR simulation outcomes (mean ± std, 30 runs each).

Topology	**Peak** I^*/N **(%)**	**Time to peak** t^*	**Final** $R(\infty)$ **(%)**
Erdős–Rényi	79.8 ± 2.8	9.6 ± 1.3	100.0 ± 0.1
Barabási–Albert	69.0 ± 3.7	11.7 ± 1.6	98.5 ± 1.0
Watts–Strogatz	59.0 ± 4.8	21.1 ± 3.0	99.7 ± 0.4
Lattice	32.6 ± 6.8	43.6 ± 18.6	89.7 ± 14.6

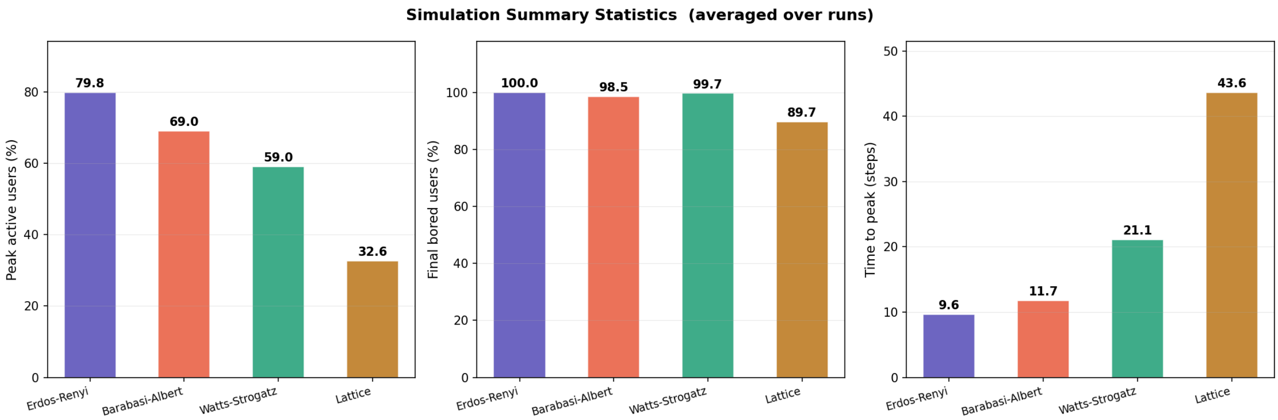

Figure 4.5: Bar chart summary of key simulation metrics by topology. Error bars are omitted for clarity; see Table 4.2 for standard deviations.

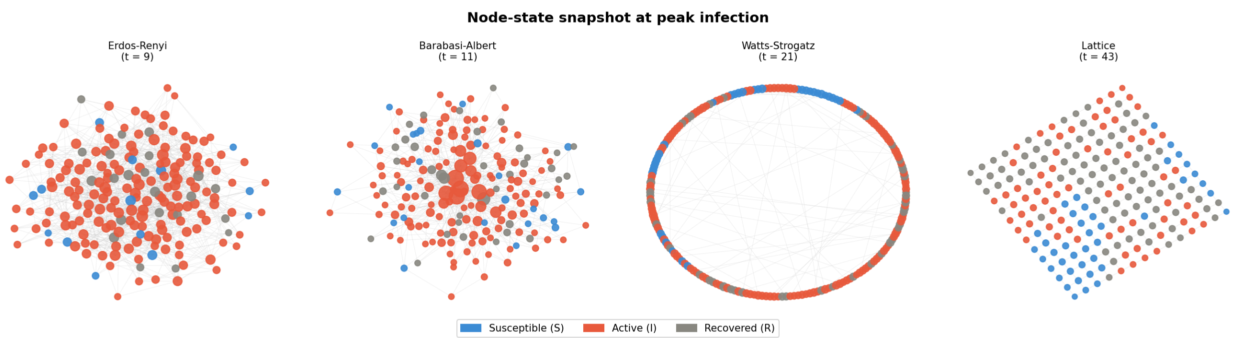

Figure 4.6: Node-state snapshot at the moment of peak infection for each topology. Blue: Susceptible; Red: Active; Grey: Recovered.

Chapter 5

Discussion and Conclusion

The results demonstrate that the topology of a social network is a primary determinant of how quickly and widely an app spreads among its users. Applying the SIR epidemic model to four canonical graph structures: Erdős–Rényi, Barabási–Albert, Watts–Strogatz, and lattice, revealed major differences in both adoption intensity and propagation speed. Peak active-user prevalence ranged from 79.8% in the ER network, reached in under 10 simulation steps, to only 32.6% in the lattice network, reached after 44 steps. These differences arise directly from structural properties: the high mean degree of ER graphs, the hub-dominated structure of BA networks, the clustering and shortcut combination of WS networks, and the absence of long-range connections in lattice topologies.

These findings have direct implications for app growth strategies. In scale-free social networks, which resemble many real-world online platforms, seeding a small number of high-degree hub users can trigger rapid adoption cascades. The BA simulations illustrate this clearly: a single hub with degree $k = 41$ can expose dozens of users simultaneously, creating super-spreading events that dramatically compress the adoption timeline.

In contrast, locally connected networks behave very differently. The lattice results suggest that adoption spreads slowly and may fail to reach distant parts of the graph due to the lack of shortcuts. In such environments, distributing multiple seed users across different geographic or social regions would likely be more effective.

The Watts–Strogatz model represents an intermediate case that may best approximate many real social systems. Strong clustering allows local communities to reinforce adoption internally, while occasional long-range links ensure eventual global spread. Since many online social networks exhibit small-world characteristics, the WS model may provide the most ecologically realistic framework among the four considered in this study.

Despite these insights, several simplifications limit the direct applicability of the model. First, the transmission and recovery parameters, β and γ, are homogeneous across all nodes and edges, whereas real users differ in influence, engagement, and willingness to recommend or uninstall applications. Second, the network topology remains static throughout the simulations, although real social networks evolve continuously as users join, leave, and reconfigure connections. Third, recovered users cannot become susceptible again; more realistic SIS or SIRS formulations would allow users to reinstall or rediscover an application after disengagement. The model also excludes external influences such as advertising campaigns, media coverage, or app-store rankings, all of which may bypass the social graph entirely. Finally, the relatively small network size ($N = 200$) introduces finite-size effects, particularly in the lattice simulations where outcome variance remains high. Larger-scale simulations would likely produce smoother threshold behaviour and more robust statistical trends.

Overall, the study demonstrates that network structure alone can fundamentally alter epidemic-style app adoption dynamics. Understanding these structural effects is therefore essential for designing efficient marketing and growth strategies in digitally connected populations.

Future work could extend the model in several directions, including heterogeneous transmission probabilities, dynamic network rewiring, SIRS-style re-infection dynamics, and multilayer social networks in which different communication channels operate simultaneously across the same population.

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